

RESEARCH PAPER ANALYSIS

Smart Community Healthcare Access Platform

Purpose of this document

This document explains the three research papers mentioned in the PBL Progress Report. For each paper, it explains what the researchers tried to solve, the main idea and method, the important findings, how the paper can help the proposed project, and the gap that remains when the paper is compared with the project requirements.

Project: Smart Community Healthcare Access Platform

Report coverage: Unit-I (Trees) and Unit-II up to Graph Representation

Project focus: efficient record organisation and representation of the community healthcare network

Research papers reviewed

Paper 1: Lapão, Correia & Jevtic (2023) — Public Health Framework for Smart Cities within the Comprehensive Approach to Sustainability in Europe: Case Study of Diabetes

Paper 2: Schuster, Polleres & Wachs (2024) — Stress-testing Road Networks and Access to Medical Care

Paper 3: Cruz Castañeda & Bertemes Filho (2024) — Improvement of an Edge-IoT Architecture Driven by Artificial Intelligence for Smart-Health Chronic Disease Management

1. Project Context

The proposed Smart Community Healthcare Access Platform is intended to help residents find healthcare facilities and to organise patient and facility information. The PBL report divides the problem into two main parts: first, storing and searching records efficiently; second, representing the network of hospitals, clinics, pharmacies and localities. The current report uses tree structures for the record-management side and graph representation for the facility-network side.

At the current stage, the project is mainly at the problem-understanding and initial design stage. The report says that a sample dataset was prepared, a BST was hand-built and traced, and an adjacency matrix and adjacency list were prepared for a small facility network. Graph traversal and shortest-path work are planned for a later stage because they were not yet covered in the current month.

2. Quick Comparison of the Three Papers

Paper	Main focus	Useful project area	Main gap for our project
1	Smart public health in smart cities and a diabetes case study	Overall system idea and digital-health context	No simple tree/graph model for community records and facilities
2	Road-network disruption and hospital accessibility in Austria	Graph view of connected locations and healthcare access	Country-scale model; not focused on simple local record management
3	Edge-IoT and AI for chronic-disease monitoring	Digital health architecture and health-data flow	Focuses on sensing/prediction, not facility search and data structures

3. Research Paper 1

Title: Public Health Framework for Smart Cities within the Comprehensive Approach to Sustainability in Europe: Case Study of Diabetes

Authors: Luís Velez Lapão, Jorge César Correia and Marija Jevtic

Journal: Sustainability, 2023, 15(5), 4269

DOI: 10.3390/su15054269

3.1 What is this paper about?

This paper looks at the role of public health in a smart city. The main idea is that a smart city should not only use technology for transport, energy or communication. It should also use digital systems to support public health. The authors propose a Smart Public Health City (SPHEC) framework and use diabetes as a case study.

The paper is therefore more about the overall design and planning of smart public-health services than about a particular data structure or algorithm. It discusses how city-level health information, digital services, sensors and decision support can work together.

3.2 Problem addressed by the paper

- Public-health problems in cities are connected with environment, lifestyle, mobility and access to services.
- Cities collect large amounts of information, but this information needs to be organised and used for better decisions.

- Health services should be included as an important part of the smart-city system.
- The authors wanted a framework that could connect public-health functions with digital tools and support city decision-making.

3.3 Approach used

The authors use Design Science Research Methodology (DSRM). In simple terms, they identify the problem, decide what a solution should provide, design the framework, demonstrate it using diabetes as a case, evaluate it and communicate the results.

The study involved experts in digital public health, smart cities, sustainability and diabetes. The framework identifies public-health functions that can be supported digitally. Examples include air-quality information, infection-disease information, health and safety information, health promotion, climate-health monitoring and early-warning functions.

3.4 Important ideas and findings

- Smart-city systems can collect and analyse health-related data and use the results to improve public-health services.
- The paper identifies digital services as an important link between public-health needs and smart-city technology.
- For infection-disease control, the framework suggests integrating infection information within city health institutions.
- The paper also discusses apps, sensors, alerts, digital information systems and chatbots for public-health activities.
- The diabetes case shows how different city factors can be considered together instead of treating healthcare as a completely separate system.

3.5 How this paper helps our project

- **Broader purpose:** our platform can be viewed as one part of a smart-community healthcare system rather than only a data-structure project.
- **Connecting residents and healthcare:** the paper supports digital services that provide useful health information to citizens.
- **Data integration:** our platform can organise facility and patient information so that different parts of the healthcare system can be accessed in a structured way.
- **Future features:** alerts and other digital-health services can be considered later if they fit the project scope.

3.6 Gap found for our project

The paper provides a high-level smart-public-health framework, but it does not give a simple implementation model for storing patient/facility records using trees or for representing a small community healthcare network using graph data structures. It is also broader than our project because it considers many public-health functions such as environment, lifestyle and policy.

Project gap: there is room for a simpler, community-level platform that takes the general smart-health idea and converts it into a practical data-organisation model using tree-based records and a graph-based facility network.

4. Research Paper 2

Title: Stress-testing Road Networks and Access to Medical Care

Authors: Hannah Schuster, Axel Polleres and Johannes Wachs

Journal: Transportation Research Part A: Policy and Practice, Volume 181, 104017, 2024

DOI: 10.1016/j.tra.2024.104017

4.1 What is this paper about?

This paper studies how road-network problems can affect people's access to hospitals, especially during crises or natural disasters. The researchers use Austria as the study area. They model the road system as a network and test what happens when important road connections are removed.

The paper is useful for understanding that a healthcare-access problem is not only about where hospitals are. It is also about how locations are connected to those hospitals. If an important road connection fails, some people may have to travel much farther or may be redirected to another hospital.

4.2 Problem addressed by the paper

- Emergency healthcare depends on transportation networks.
- A road disruption can increase travel time to a hospital.
- Some road connections are much more important than others.
- A disruption can change which hospital becomes the nearest hospital for a population.
- Hospitals may receive a sudden increase in patients when other routes are disrupted.

4.3 Approach used

The researchers start with a very detailed Austrian road network and simplify it into a smaller network. In the simplified model, municipalities are treated as nodes and road connections between municipalities are treated as corridors/edges. This makes repeated network simulations more practical.

They perform stress tests by removing one corridor at a time and also by removing a corridor together with nearby corridors. They measure how much hospital accessibility changes. One measure is the Accessibility Corridor Impact Score (ACIS), which ranks how strongly the removal of a corridor affects access to hospitals.

4.4 Important findings

- Most road-corridor closures have a relatively small effect, but a small number of corridors have a very large effect.
- Some highly important corridors contain only a few roads, meaning that they may have limited redundancy.
- The study identifies municipalities that become more vulnerable when important connections are removed.
- A corridor closure can change hospital catchment areas, causing some hospitals to receive more patients.
- The study shows that network structure can be used to identify critical healthcare-access connections.

4.5 How this paper helps our project

- **Graph idea:** hospitals, clinics, pharmacies and localities can be treated as nodes, while roads or connections can be represented as edges.
- **Why connections matter:** simply storing a list of hospitals is not enough when the project also wants to represent how facilities are connected.
- **Future direction:** after graph traversal/shortest-path concepts are covered, the project can study which facility can be reached more easily.
- **Graph information:** distances or connection weights can later be included in the graph.

4.6 Gap found for our project

The paper studies a very large road and healthcare network for the whole of Austria. It uses detailed geographic data, simulations and specialised accessibility measures. Our project has a much smaller and simpler goal: representing a community-level healthcare network and combining it with patient/facility record management.

Project gap: there is a need for a simpler graph model that a community-level healthcare platform can use to represent nearby hospitals, clinics, pharmacies and localities, without requiring a country-scale GIS and simulation system.

5. Research Paper 3

Title: Improvement of an Edge-IoT Architecture Driven by Artificial Intelligence for Smart-Health Chronic Disease Management

Authors: William Alberto Cruz Castañeda and Pedro Bertemes Filho

Journal: Sensors, 2024, 24(24), 7965

DOI: 10.3390/s24247965

5.1 What is this paper about?

This paper proposes an AI-driven edge-IoT architecture for smart-health and chronic-disease management. The main example is non-invasive glucose prediction. The system collects physiological information using a wearable device, performs some processing close to the data source, sends the information through an edge node, and uses a cloud learning layer for AI-based analysis.

The paper therefore focuses on how health data can be collected, processed and analysed in a connected smart-health system. It is different from our project because it is mainly concerned with chronic-disease monitoring and AI prediction rather than finding healthcare facilities.

5.2 Main architecture

- **Edge Device Layer:** a wearable device collects physiological data such as oxygen concentration, pulse rate, skin temperature and bio-impedance.
- **Edge Node Layer:** receives the data, performs nearby computing/caching and converts the data into a standard JSON format before sending it onward.
- **Cloud Learning Layer:** handles the data, trains an AI model and performs inference on new input data.

5.3 AI case study and results

The authors test the architecture using non-invasive glucose prediction. They compare twelve regression models using measures such as mean squared error (MSE), root mean squared error (RMSE) and R^2 . The paper reports that the CatBoost Regressor performed best among the tested models in the reported case study.

The study is presented as a practical demonstration of how wearable data, edge processing and cloud AI can be combined for chronic-disease management. The authors also identify future work such as larger datasets, embedded AI models, wearable deployment, an autonomous AI cloud engine and federated learning.

5.4 How this paper helps our project

- **Smart-health direction:** our platform can be part of a wider digital healthcare ecosystem.
- **Data flow:** healthcare data can be collected, organised, transferred and analysed in different layers.
- **Future idea:** if the project later includes health-monitoring data, local processing and cloud-based analysis can be considered.
- **Organised health data:** our record-management part can provide a structured way of storing healthcare information before future analysis.

5.5 Gap found for our project

The paper mainly deals with chronic-disease monitoring, wearable sensors, edge computing and AI prediction. It does not focus on helping a resident find a suitable nearby healthcare facility, nor does it provide a tree-based model for patient/facility records or a simple graph model for hospitals, clinics and pharmacies.

Project gap: the proposed platform can focus on the community healthcare-access side and combine structured record management with a facility-network model. This complements the smart-health monitoring architecture discussed in the paper.

6. Combined Learning from the Three Papers

The three papers approach healthcare from different levels. Paper 1 gives a broad smart-city and public-health view. Paper 2 focuses on the network that connects people to hospitals. Paper 3 focuses on collecting and analysing health data through an edge-IoT and AI architecture. Together, they helped identify three useful ideas for the proposed project.

Paper	What we learned	Project connection
Paper 1	Healthcare can be treated as an important part of a smart community/city and digital services can support public health.	Provides the overall motivation and smart-health context.
Paper 2	Healthcare accessibility depends on how locations and hospitals are connected.	Supports the graph representation of facilities and localities.
Paper 3	Health information can be collected and processed through connected digital layers and AI.	Provides future direction for organised health data and possible smart-health integration.

7. Final Research Gap Identified

After comparing the three papers with the proposed Smart Community Healthcare Access Platform, the main gap identified is not that previous research is missing healthcare technology. Instead, the gap is the absence of a simple, community-level and data-structure-oriented solution that combines the following parts in one platform:

- Efficient organisation and searching of patient and healthcare-facility records using suitable tree structures.
- Representation of hospitals, clinics, pharmacies and localities as a graph using a simple adjacency matrix or adjacency list.
- A single platform that connects record management with the healthcare-facility network.
- A design that is easier to understand and implement at community level, instead of depending on country-scale GIS, complex simulations, wearable hardware or a large smart-city infrastructure.
- A base on which later search/traversal features can be added after the required DSA topics are covered.

This gap fits the current PBL work because the project has already separated the problem into record management and network modelling. The current report also states that a sample BST and graph representations have been prepared, while more advanced graph-search work is planned for a later stage.

8. How the Research Papers Support the Project Development

- **Step 1 – Problem understanding:** Paper 1 helped establish why healthcare access and digital public-health services matter in a smart community.
- **Step 2 – Network modelling:** Paper 2 showed why the connections between locations and healthcare facilities are important, supporting the graph part of the project.
- **Step 3 – Health-data direction:** Paper 3 showed how health data can be collected, processed and analysed in a connected smart-health system.
- **Step 4 – Project gap:** the project focuses on a simpler combination of record organisation and facility-network representation that is not the main focus of any of the three papers.
- **Step 5 – Future development:** later reviews can extend the current model with graph traversal, facility searching and other features once the corresponding DSA topics are covered.

9. Conclusion

The three research papers provide useful background from three different viewpoints. The first paper explains the broader role of digital public health in smart cities. The second shows how network structure affects access to hospitals. The third shows how connected health devices, edge computing and AI can support chronic-disease management. None of them directly provides the simple tree-plus-graph model proposed for the Smart Community Healthcare Access Platform. Therefore, the project can focus on this practical gap by organising records efficiently and representing the local healthcare network in a clear and manageable form.

The purpose of the literature review is not to copy the methods of these papers. Instead, the papers help us understand the problem, identify useful directions and clearly define what our project will try to add.

10. References

1. Lapão, L. V., Correia, J. C., & Jevtic, M. (2023). Public Health Framework for Smart Cities within the Comprehensive Approach to Sustainability in Europe: Case Study of Diabetes. *Sustainability*, 15(5), 4269. DOI: 10.3390/su15054269.
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3. Cruz Castañeda, W. A., & Bertemes Filho, P. (2024). Improvement of an Edge-IoT Architecture Driven by Artificial Intelligence for Smart-Health Chronic Disease Management. *Sensors*, 24(24), 7965. DOI: 10.3390/s24247965.